

IJAY NARANG

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EDUCATION

Georgia Institute of Technology Atlanta, GA	2025-2028*
• Ph.D. Candidate in Computer Science	
• M.S. in Machine Learning	2025-2026*
Princeton University Princeton, NJ GPA 3.9/4.0	2021-2025
• B.S.E. Computer Science, summa cum laude, with minors in Applied and Computational Mathematics, Finance, and Optimization and Quantitative Decision Science	
Ardrey Kell High School Charlotte, NC GPA 4.0/4.0	2017-2021
• U.S. Presidential Scholar, four-time AIME Qualifier, two-time National Chemistry Olympiad Finalist, Valedictorian (1/814)	

EXPERIENCE

Princeton Vision & Learning Lab Visiting Researcher	Summer '26
• Developed GlimpseBC, which trains a policy to select sequential visual glimpses using demonstrations generated with full-image access; work submitted to ICLR.	
Princeton University Computer Science Department Course Developer	Summer '25
• Developed course curriculum and wrote lecture notes for Princeton's new intermediate theoretical computer science class, COS 330.	
Capula Investment Management Trading and Research Intern	Summer '24
• Built a monitoring system for correlations and dislocations between CDS spreads. Developed a backtester and used it to evaluate and optimize relative-value trading strategies.	
Siemens Digital Industries Software Development Intern	Summers '22 and '23
• Implemented frontend components displaying client policies for the Operations Insights Dashboard with React.js.	
• Developed an invoice generator using PostgreSQL for the Digital E-Commerce billing module.	
Solar Operations Solutions, LLC Data Science Intern	Summer '21
• Used linear time-series models to forecast energy demand and regulate supply across solar sites.	

SELECTED WORK

(In Submission to ICLR '27) **GlimpseBC: Learning Saccadic Visual Policies from Global View Demonstrations** (with Jia Deng)

(ICML '26) **Steady-State Behavior of Constant-Stepsize Stochastic Approximation: Gaussian Approximation and Tail Bounds** (with Zedong Wang, Yuyang Wang, Felix Wang, Yuzhou Wang, and Siva Theja Maguluri) [\[View Paper Here\]](#)

(SODA '27) **Computational Thresholds for Balanced and Fixed-Slice Independent Sets in Bipartite Graphs** (with Will Perkins, Yuzhou Wang, and Timothy L. H. Wee) [\[View Paper Here\]](#)

(In Submission to ALT'27) **Optimal Detection of Planted Stars via a Random Energy Model** (with Will Perkins and Timothy Wee) [\[View Paper Here\]](#)

(In submission to SICON) **Complexity of Output Feedback Stabilization** (with Amir Ali Ahmadi, Abraar Chaudhry, and Yukai Tang) [\[View Paper Here\]](#)

(Preprint) **Improved SDP Coloring of 3-Colorable Graphs from Recursive Gaussian Certificates** (with Yukai Tang) [\[View Paper Here\]](#)

(Preprint) **Structural Corrections to the Bethe Approximation of the Permanent** (with Will Perkins) [\[View Paper Here\]](#)

(Preprint) **The Hard-Core Model on Bipartite Spectral Expanders: Counting and Sampling at All Fugacities** (with Will Perkins) [\[View Paper Here\]](#)

* Expected graduation dates.

(Preprint) **Schrijver Number Quasi-Tensorization and Multicolor Ramsey Bounds via Robust OR Polynomials** (with Yukai Tang) [\[View Paper Here\]](#)

AWARDS AND HONORS

- Georgia Tech Presidential Fellowship (\$5,000 Award)
- Computer Science Outstanding Student Teaching Award, Princeton University (\$1,000 Award)
- Outstanding Computer Science Research Prize, Princeton University (\$1,500 Award)
- Computer Science Service Award, Princeton University
- Highest Honors, Princeton Computer Science Department

SKILLS AND PROJECTS

- **Programming Languages:** Python, OCaml, R, Java, SQL, JavaScript, HTML, and CSS.
- **Compass:** Built an all-in-one Princeton University course-planning app, now used by 700+ students and maintained by Hoagie Club as Hoagie Plan. [\[See the App Here\]](#)
- **Chess:** Achieved a Second Category title with a peak USCF rating of 1902.